

Qifan Zhang

Senior Staff Researcher · Palo Alto Networks

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RESEARCH PROFILE

I study the security of Internet protocols and the systems that rely on them. My work combines fuzzing, protocol analysis, and formal methods to find vulnerabilities in widely deployed software. My research spans DNS and DNSSEC security, automated vulnerability discovery, and security and privacy in machine learning.

APPOINTMENT

Palo Alto Networks — Senior Staff Researcher Jan 2025–present

EDUCATION

University of California, Irvine

Ph.D. in Computer Engineering Sep 2020–Mar 2025

M.S. in Computer Engineering Sep 2020–Jun 2022

Advisor: Prof. Zhou Li · Department of Electrical Engineering and Computer Science

ShanghaiTech University

Aug 2016–Jul 2020

B.Eng. in Computer Science and Technology

Minor in Innovation and Entrepreneurship

University of California, Berkeley — Summer Session 2017

RESEARCH HIGHLIGHTS

- **ResolverFuzz.** Stateful, grammar-based and differential fuzzing of DNS resolvers. The study identified **23 vulnerabilities** across six DNS implementations, resulting in **15 CVEs** [C5].
- **DNS caching and revocation.** Coauthored MaginotDNS (**3 CVEs**) and Phoenix Domain (**9 CVEs**) [C3, C2]. MaginotDNS earned a \$1,000 Microsoft Security Response Center award.
- **Model extraction.** Demonstrated extraction of autonomous-driving localization models with centimeter-level accuracy using short driving traces [C1].
- **Federated learning security.** Contributed to the FedSecurity attack/defense benchmark and CustodianFL, which verifies anomaly detection with zero-knowledge proofs [C4, W1].

TECHNICAL EXPERTISE

Languages: Python, C/C++, Rust, Java.

Security: Protocol analysis, stateful and grammar-based fuzzing, differential testing, DNS measurement, Scapy, GDB, CLion.

Systems and ML: Docker, VMware, PyTorch, MATLAB/Simulink, OpenCV, Baidu Apollo, Cloudflare API.

PUBLICATIONS

Conference papers

- [C6] **Mind the Gap: Cross-RFC Formal Analysis Exposes DNSSEC's Hidden Vulnerabilities**
Qifan Zhang, Zilin Shen, Imtiaz Karim, Elisa Bertino, Daiping Liu, and Zhou Li.
ACM CCS 2026. ACM SIGSAC Conference on Computer and Communications Security. [Code](#)
- [C5] **ResolverFuzz: Automated Discovery of DNS Resolver Vulnerabilities with Query-Response Fuzzing**
Qifan Zhang, Xuesong Bai, Xiang Li, Haixin Duan, Qi Li, and Zhou Li.
USENIX Security 2024. 33rd USENIX Security Symposium. [Extended version](#) · [Code](#)
- [C4] **FedSecurity: A Benchmark for Attacks and Defenses in Federated Learning and Federated LLMs**
Shanshan Han, Baturalp Buyukates, Zijian Hu, Han Jin, Weizhao Jin, Lichao Sun, Xiaoyang Wang, Wenxuan Wu, Chulin Xie, Yuhang Yao, Kai Zhang, **Qifan Zhang**, Yuhui Zhang, Carlee Joe-Wong, Salman Avestimehr, and Chaoyang He.
KDD 2024. 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. [Preprint](#) · [Code](#)
- [C3] **The Maginot Line: Attacking the Boundary of DNS Caching Protection**
Xiang Li, Chaoyi Lu, Baojun Liu, **Qifan Zhang**, Zhou Li, Haixin Duan, and Qi Li.
USENIX Security 2023. 32nd USENIX Security Symposium. [Project](#)
- [C2] **Ghost Domain Reloaded: Vulnerable Links in Domain Name Delegation and Revocation**
Xiang Li, Baojun Liu, Xuesong Bai, Mingming Zhang, **Qifan Zhang**, Zhou Li, Haixin Duan, and Qi Li.
NDSS 2023. 30th Network and Distributed System Security Symposium. [DOI](#)
- [C1] **Play the Imitation Game: Model Extraction Attack against Autonomous Driving Localization**
Qifan Zhang, Junjie Shen, Mingtian Tan, Zhe Zhou, Zhou Li, Qi Alfred Chen, and Haipeng Zhang.
ACSAC 2022. 38th Annual Computer Security Applications Conference.

Workshop paper

- [W1] **Kick Bad Guys Out! Conditionally Activated Anomaly Detection in Federated Learning with Zero-Knowledge Proof Verification**
Shanshan Han, Wenxuan Wu, Baturalp Buyukates, Weizhao Jin, **Qifan Zhang**, Yuhang Yao, and Salman Avestimehr.
PRISM 2026. Workshop on Attack Provenance, Reasoning, and Investigation for Security in the Monitored Environment, co-located with NDSS. · [DOI](#)

Journal paper

- [J1] **A Comprehensive Study of DNS Operational Issues by Mining DNS Forums**
Xianran Liao, Jiachen Xu, **Qifan Zhang**, and Zhou Li.
IEEE Access 2022. Vol. 10, pp. 110807–110820.

ACADEMIC SERVICE

Program committees. USENIX Security (2027); ACSAC (2026, 2025); SiMLA (2025).
Artifact evaluation. NDSS (2025, 2024); ACM CCS (2024, 2023); USENIX Security (2024).
External conference reviewing. NDSS (2025, 2023, 2022, 2021); AsiaCCS (2022, 2021); SecureComm (2023).
Journal reviewing. IEEE Transactions on Dependable and Secure Computing; IEEE Transactions on Information Forensics and Security; IEEE Transactions on Wireless Communications; IEEE Internet of Things Journal; Computer Networks; High-Confidence Computing; Peer-to-Peer Networking and Applications; PeerJ Computer Science.

PRESENTATIONS

ResolverFuzz — USENIX Security, Philadelphia	Aug 2024
ResolverFuzz — NDSS poster session, San Diego	Feb 2024
ResolverFuzz — DNS-OARC 42, Charlotte	Feb 2024
Invited seminar — online	Oct 2023
Play the Imitation Game — ACSAC, Austin	Dec 2022

TEACHING

University of California, Irvine

- **EECS 148: Computer Networks.** Head TA, Spring 2024; 217 students.
- **EECS 40: Object-Oriented System & Programming.** TA, Fall 2022 and 2023 (95 and 90 students); head TA in Fall 2023.

ShanghaiTech University

- **SI 100B: Introduction to Information Science and Technology.** Core TA, Spring 2018, 2019, and 2020; 203, 174, and 410 students, respectively.
- **CS 277: Introduction to Data Science and FinTech.** TA, Fall 2019; 23 students.
- **CS 100: Programming.** TA, Fall 2018; 243 students.
- **SI 100C: Introduction to Computer Science and Technology.** TA, Fall 2017; 127 students.

HONORS AND AWARDS

UCI Concerto Competition — winner	2024
ShanghaiTech Outstanding Graduate	2020
SIST Outstanding Teaching Assistant , ShanghaiTech	2020, 2019, 2018
Merit Student , ShanghaiTech	2018–19, 2017–18, 2016–17
Outstanding Personnel , ShanghaiTech Summer Camp	2017

Research travel support. USENIX Security (2024, 2021); ANRW (2023); ACSAC Student Conferenceship (2022); NDSS (2021); IEEE Symposium on Security and Privacy (2021).
UCI Associated Graduate Students Conference Stipend. Winter 2024 and Winter 2022.